

Assignment 2

00103335: Deep Learning and Reinforcement Learning

Fall 2026

All norms are Euclidean. No assumptions beyond those in the questions are needed; in particular, no second derivatives or Lipschitz gradient assumptions are used.

Problem 1

(i) $\Rightarrow$ (ii). For $0 < t < 1$, convexity gives

$$f(x + t(y - x)) = f((1 - t)x + ty) \leq (1 - t)f(x) + tf(y).$$

Subtract $f(x)$ and divide by $t > 0$:

$$\frac{f(x + t(y - x)) - f(x)}{t} \leq f(y) - f(x).$$

Differentiability gives

$$\nabla f(x)^\top (y - x) = \lim_{t \downarrow 0} \frac{f(x + t(y - x)) - f(x)}{t} \leq f(y) - f(x),$$

which is (ii).

(ii) $\Rightarrow$ (i). Let $z = \lambda x + (1 - \lambda)y$. Apply (ii) with base point z and endpoints x, y , multiply by $\lambda, 1 - \lambda$, and add:

$$\lambda f(x) + (1 - \lambda)f(y) \geq f(z) + \nabla f(z)^\top [\lambda(x - z) + (1 - \lambda)(y - z)] = f(z).$$

(ii) $\Rightarrow$ (iii). Write (ii) in both directions:

$$f(y) - f(x) \geq \nabla f(x)^\top (y - x), \quad f(x) - f(y) \geq \nabla f(y)^\top (x - y).$$

Adding cancels the function values and gives

$$0 \geq \nabla f(x)^\top (y - x) + \nabla f(y)^\top (x - y),$$

which is (iii).

(iii) $\Rightarrow$ (ii). For $x \neq y$, set $h = y - x$ and $g(t) = f(x + th)$. The mean value theorem gives $g(1) - g(0) = g'(c)$ for some $c \in (0, 1)$. By (iii) at $x, x + ch$,

$$(\nabla f(x + ch) - \nabla f(x))^\top (ch) \geq 0, \quad g'(c) \geq \nabla f(x)^\top h.$$

This proves (ii); for $x = y$ it is equality.

Problem 2

Define $g(x) = f(x) - \frac{\mu}{2} \|x\|^2$, so $\nabla g(x) = \nabla f(x) - \mu x$.

(i) $\Leftrightarrow$ (ii). Set $z = \alpha x + (1 - \alpha)y$. The convexity inequality for g is

$$f(z) - \frac{\mu}{2} \|z\|^2 \leq \alpha f(x) + (1 - \alpha)f(y) - \frac{\mu}{2} (\alpha \|x\|^2 + (1 - \alpha) \|y\|^2).$$

The quadratic difference expands as

$$\begin{aligned} \alpha \|x\|^2 + (1 - \alpha) \|y\|^2 - \|z\|^2 &= (\alpha - \alpha^2) \|x\|^2 + ((1 - \alpha) - (1 - \alpha)^2) \|y\|^2 \\ &\quad - 2\alpha(1 - \alpha)x^\top y \\ &= \alpha(1 - \alpha) (\|x\|^2 + \|y\|^2 - 2x^\top y) \\ &= \alpha(1 - \alpha) \|y - x\|^2. \end{aligned}$$

Moving $\mu \|z\|^2 / 2$ to the right therefore gives

$$f(z) \leq \alpha f(x) + (1 - \alpha)f(y) - \frac{\mu}{2} \alpha(1 - \alpha) \|y - x\|^2,$$

which is (ii). All the algebraic steps are reversible.

(i) $\Leftrightarrow$ (iii). By Problem 1, convexity of g is equivalent to $g(y) \geq g(x) + \nabla g(x)^\top (y - x)$. Substituting the definitions gives

$$f(y) - \frac{\mu}{2} \|y\|^2 \geq f(x) - \frac{\mu}{2} \|x\|^2 + (\nabla f(x) - \mu x)^\top (y - x).$$

Rearranging yields

$$\begin{aligned} f(y) &\geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2} (\|y\|^2 - \|x\|^2 - 2x^\top (y - x)) \\ &= f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2} \|y - x\|^2, \end{aligned}$$

because the parenthesis equals $\|y\|^2 + \|x\|^2 - 2x^\top y = \|y - x\|^2$. Reversing these steps proves the converse.

(i) $\Leftrightarrow$ (iv). Again by Problem 1, convexity of g is equivalent to

$$0 \leq (\nabla g(y) - \nabla g(x))^\top (y - x) = (\nabla f(y) - \nabla f(x))^\top (y - x) - \mu \|y - x\|^2,$$

which is (iv). Thus all four conditions are equivalent.

Problem 3

Use the numbering in the question. Write $h = y - x$ and $a = \nabla f(y) - \nabla f(x)$.

(ii) Problem 2(iv) and Cauchy-Schwarz give

$$\mu \|h\|^2 \leq a^\top h \leq \|a\| \|h\|.$$

For $h \neq 0$, divide by $\|h\|$ to obtain $\|a\| \geq \mu \|h\|$; the result is immediate for $h = 0$.

(iii) Apply strong convexity with base point y :

$$f(x) \geq f(y) + \nabla f(y)^\top (x - y) + \frac{\mu}{2} \|h\|^2.$$

Rearranging and completing the square gives

$$\begin{aligned} f(y) - f(x) - \nabla f(x)^\top h &\leq a^\top h - \frac{\mu}{2} \|h\|^2 \\ &= \frac{1}{2\mu} \|a\|^2 - \frac{\mu}{2} \|h - a/\mu\|^2 \leq \frac{1}{2\mu} \|a\|^2. \end{aligned}$$

(iv) By Cauchy–Schwarz and (ii),

$$a^\top h \leq \|a\| \|h\| \leq \frac{1}{\mu} \|a\|^2.$$

Problem 4

(a) Strong convexity at x_t implies

$$f(x^*) \geq f(x_t) + \nabla f(x_t)^\top (x^* - x_t) + \frac{\mu}{2} d_t^2.$$

Rearrangement gives $\nabla f(x_t)^\top (x_t - x^*) \geq \Delta_t + \frac{\mu}{2} d_t^2$.

(b) Expanding the update and using (a) and the stated bound,

$$\begin{aligned} d_{t+1}^2 &= d_t^2 - 2\eta_t \nabla f(x_t)^\top (x_t - x^*) + \eta_t^2 \|\nabla f(x_t)\|^2 \\ &\leq (1 - \mu\eta_t) d_t^2 - 2\eta_t \Delta_t + \eta_t^2 L^2. \end{aligned}$$

(c) Substitute $\eta_t = 2/[\mu(t+1)]$ into (b):

$$d_{t+1}^2 \leq \frac{t-1}{t+1} d_t^2 - \frac{4}{\mu(t+1)} \Delta_t + \frac{4L^2}{\mu^2(t+1)^2}.$$

Multiplying by $\mu t(t+1)/4$ and rearranging yields

$$t\Delta_t \leq \frac{\mu}{4} [t(t-1)d_t^2 - t(t+1)d_{t+1}^2] + \frac{tL^2}{\mu(t+1)}.$$

To sum, shift the index $s = t+1$ in the negative term:

$$\begin{aligned} &\sum_{t=1}^T t(t-1)d_t^2 - \sum_{t=1}^T t(t+1)d_{t+1}^2 \\ &= \sum_{t=1}^T t(t-1)d_t^2 - \sum_{s=2}^{T+1} s(s-1)d_s^2 = -T(T+1)d_{T+1}^2. \end{aligned}$$

The terms with indices $2, \dots, T$ cancel and the initial coefficient $1(1-1)$ is zero. Thus

$$\sum_{t=1}^T t\Delta_t \leq -\frac{\mu}{4} T(T+1)d_{T+1}^2 + \frac{L^2}{\mu} \sum_{t=1}^T \frac{t}{t+1} \leq \frac{TL^2}{\mu}.$$

(d) The weights $w_t = 2t/[T(T+1)]$ are nonnegative and satisfy

$$\sum_{t=1}^T w_t = \frac{2}{T(T+1)} \sum_{t=1}^T t = \frac{2}{T(T+1)} \frac{T(T+1)}{2} = 1.$$

Since $\bar{x}_T = \sum_{t=1}^T w_t x_t$, convexity gives $f(\bar{x}_T) \leq \sum_{t=1}^T w_t f(x_t)$. Subtracting $f(x^*) = \sum_{t=1}^T w_t f(x^*)$ and using (c),

$$f(\bar{x}_T) - f(x^*) \leq \sum_{t=1}^T w_t \Delta_t = \frac{2}{T(T+1)} \sum_{t=1}^T t \Delta_t \leq \boxed{\frac{2L^2}{\mu(T+1)}} = O(T^{-1}).$$